

JECRC UNIVERSITY, JAIPUR
Session: Jan-June 2026
Subject: Mathematics-II (DMA025A)
B. Tech. II Semester I Year 2025-26
Tutorial Sheet-4 (M.M: 64)

UNIT-2: Ordinary Differential Equations

CO2: Understand the linear ordinary differential equations along with their applications.

PART – A (5*2=10)

A1: Define first order first degree ordinary differential equations.

A2: Find the order and degree of $p^2 + 2py \cot x = y^2$. **Ans:** 1, 2

A3: Solve the differential equation $y = px + a/p$. **Ans:** $y = cx + a/c$.

A4: Solve the differential equation $p^2 - 2p - 3 = 0$. **Ans:** $(x - y - c_1)(y - 3x - c_2) = 0$.

A5: What is Clairaut's equation?

PART – B (3*7=21)

B1: Solve the differential equation $x^2 p^2 - 2xyp + 2y^2 - x^2 = 0$. **Ans:** $\sin^{-1}(y/x) = \log cx$.

B2: Solve the differential equation $y + px = p^2 x^4$. **Ans:** $xy = c^2 x - c$.

B3: Solve the differential equation $y = 2px + y^2 p^3$. **Ans:** $y^2 = 2cx + c^3$.

PART – C (3*11=33)

C1: Solve the differential equation $p^2 + 2py \cot x = y^2$. **Ans:** $(y - c_1/(1 + \cos x))(y - c_2/(1 - \cos x))$.

C2: Solve the differential equation $p^3 - 4xyp + 8y^2 = 0$. **Ans:** $c(4x - c)^2 = 64y$.

C3: Solve the differential equation $(x - a)p^2 + (x - y)p - y = 0$. **Ans:** $y = cx - ac^2/(c + 1)$.
